

Totalizer MAPS Template (Instrumentation)

The Totalizer MAPS template is designed to control a totalizer that is connected with digital inputs that increments and decrements the totalizer accumulated value.

All MAPS templates provide the following:

- an associated PLC function block
- a set of SCADA tags (Adroit agents) linked to the provided signals
- an associated SCADA graphic with inbuilt faceplates

This document describes the Advanced Totalizer MAPS template. For details of its required PLC memory usage; number of PLC I/O and SCADA size (scan points) - see the **Resources** table below.

Advanced Totalizer Functional Philosophy

The Totalizer is accumulated and alarmed according to the setpoints from the Setup faceplate.

Field inputs

- Increment Totalizer Input
- Decrement Totalizer Input
- Reset Totalizer Input
- Enable Interlock Input

PLC Setpoints

- High-high Alarm
- High Alarm
- Low Alarm
- Low-low Alarm
- Under Range
- Over Range

Simulation mode - will override input to the Totalizer and simulate an analog in the PLC program that is controlled from the faceplate. Simulation mode is used to test the process logic without receiving the physical input itself. It is very useful in testing, commissioning and maintenance.

Alarming – The totalizer block has four alarm setpoints, each of which can be enabled independently. The high warning alarm and high-high alarm will trigger once the live value exceeds the alarm setpoint value and the low warning alarm and low-low alarm will trigger once the live value falls below the alarm setpoint value. The alarms will reset once the process value is out of the alarm range and the operator has acknowledged the alarm with the reset button on the faceplate.

Typical Faceplate Graphics for the Advanced Totalizer MAPS Template

Totalizer	
	

 0000_ABCDE_123	Tag name				
 Total 12345	Current live total				
 Last 12345	Last total on reset				
	Fault active				
	Simulation mode				
Signal Description		Agent Type	Advanced	Standard	Basic
SCADA Control Words:					
SSCL 0	Control Word	Marshall	x		
SSCL 1	Total Min	Analog	x		
SSCL 2	Total Max	Analog	x		
SSCL 3	Reset SP	Analog	x		
SSCL 4	Increment SP	Analog	x		
SSCL 5	Decrement SP	Analog	x		
SSCL 6	Low-low Alarm	Analog	x		
SSCL 7	Low Alarm	Analog	x		
SSCL 8	High Alarm	Analog	x		
SSCL 9	High-high Alarm	Analog	x		

Signal Description		Agent Type	Advanced	Standard	Basic
SCADA Status Words:					
SSSL 0	Status Word	Marshall	x		
SSSL 1	Totalizer Live Value	Analog	x		
SSSL 2	Totalizer Last Value	Analog	x		

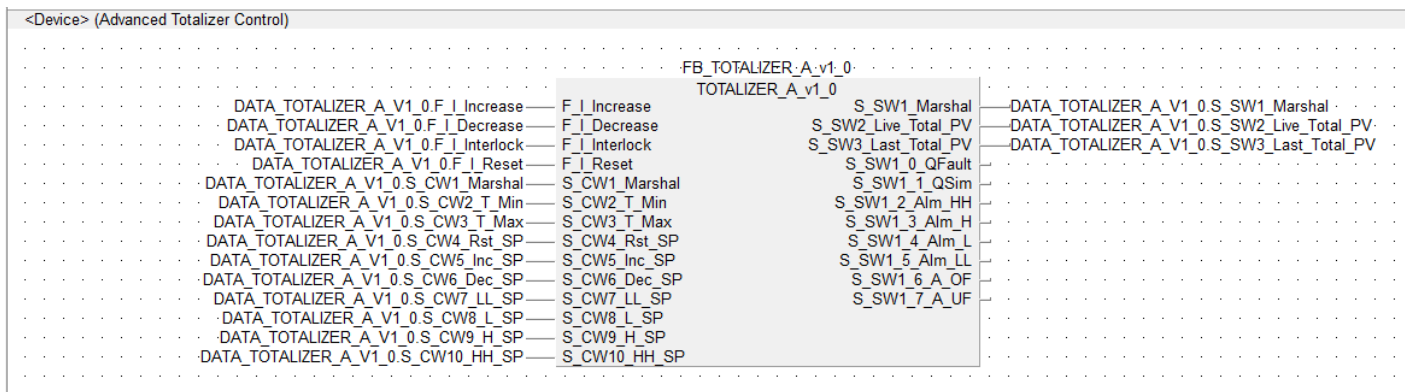
SCADA Control Word (Adroit Marshall Agent)		Event Logged	Advanced	Standard	Basic
Bit 0	High-high AI Alarm Enabled	x	x		
Bit 1	High AI Alarm Enabled	x	x		
Bit 2	Low AI Alarm Enabled	x	x		
Bit 3	Low-Low AI Alarm Enabled	x	x		
Bit 4	Increase Pulse	x	x		
Bit 5	Decrease Pulse	x	x		
Bit 6	Enable Auto Reset	x	x		
Bit 7	Enable Live Reset	x	x		
Bit 8	Enable Last Reset	x	x		
Bit 9	Spare				
Bit 10	Spare				
Bit 11	Spare				
Bit 12	Scada Simulation	x	x		
Bit 13	Fault Reset		x		
Bit 14	Data Reset	x	x		
Bit 15	Spare				

SCADA Status Word (Adroit Marshall Agent)		Alarmed	Advanced	Standard	Basic
Bit 0	Fault	x	x		
Bit 1	Simulation On	x	x		
Bit 2	High-high Alarm Active	x	x		
Bit 3	High Alarm Active	x	x		
Bit 4	Low Alarm Active	x	x		
Bit 5	Low-low Alarm Active	x	x		
Bit 6	Over Range Active	x	x		
Bit 7	Under Range Active	x	x		
Bit 8	Spare				
Bit 9	Spare				
Bit 10	Spare				
Bit 11	Spare				
Bit 12	Spare				
Bit 13	Spare				
Bit 14	Spare				
Bit 15	Spare				

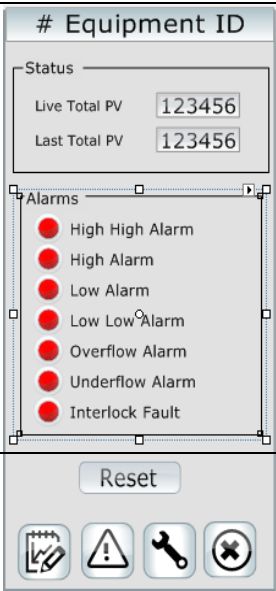
Digital Inputs:		Advanced	Standard	Basic
DI 0	Increase	x		
DI 1	Decrease	x		
DI 2	Interlock	x		
DI 3	Reset Total	x		

Resources	Advanced	Standard	Basic
Digital Inputs	4		
Digital Outputs	0		
Analog Inputs	0		
Analog Outputs	0		
PLC Memory Steps	214		
System Words Used	16		
System Bits Used	133		
SCADA SCAN Points	13		

Template: Advanced Totalizer PLC Function Block

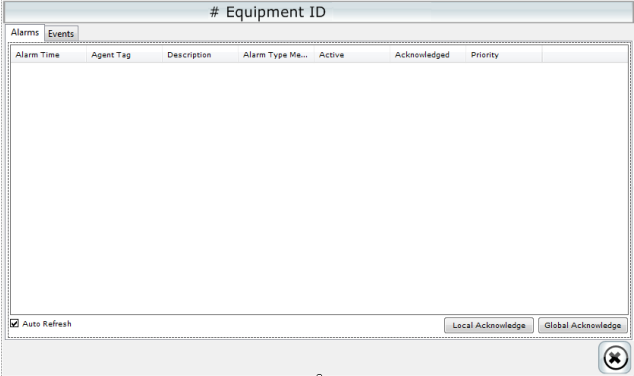


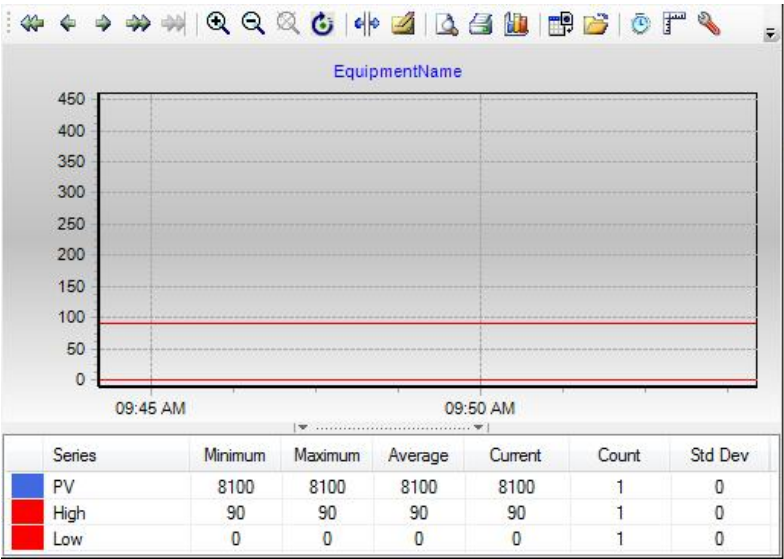
Template Graphic: Advanced Home Screen



Control	Descriptions
# Equipment ID	Totalizer controlled name
Live Total PV	Current total of totalizer
Last Total PV	Last total saved when reset occurred
High High	High-high alarm indication
High	High alarm indication
Low	Low alarm indication
Low Low	Low-low alarm indication
Overflow	over range alarm

	 Underflow	under range alarm
	 Interlock Fault	Alarm fault latch
	Reset	Reset the Totalizer after a fault has been repaired

Template Graphic: Advanced Alarms & Events Screen	Control	Descriptions
	Alarms	Display all the alarms filtered for the current Digital Output
	Events	Display all the events filtered for the current Digital Output.
	Local Acknowledge	Acknowledge all the alarms on locally on the current machine
	Global Acknowledge	Acknowledge all the alarms on globally on all the machines.

Template Graphic: Advanced Trend Screen	Control	Descriptions
	Equipment Name	Display the name of the current analog Input of the trend
	PV	Display the level of the current level
	High	High Setpoint for analog scaled value
	Low	Low Setpoint for analog scaled value